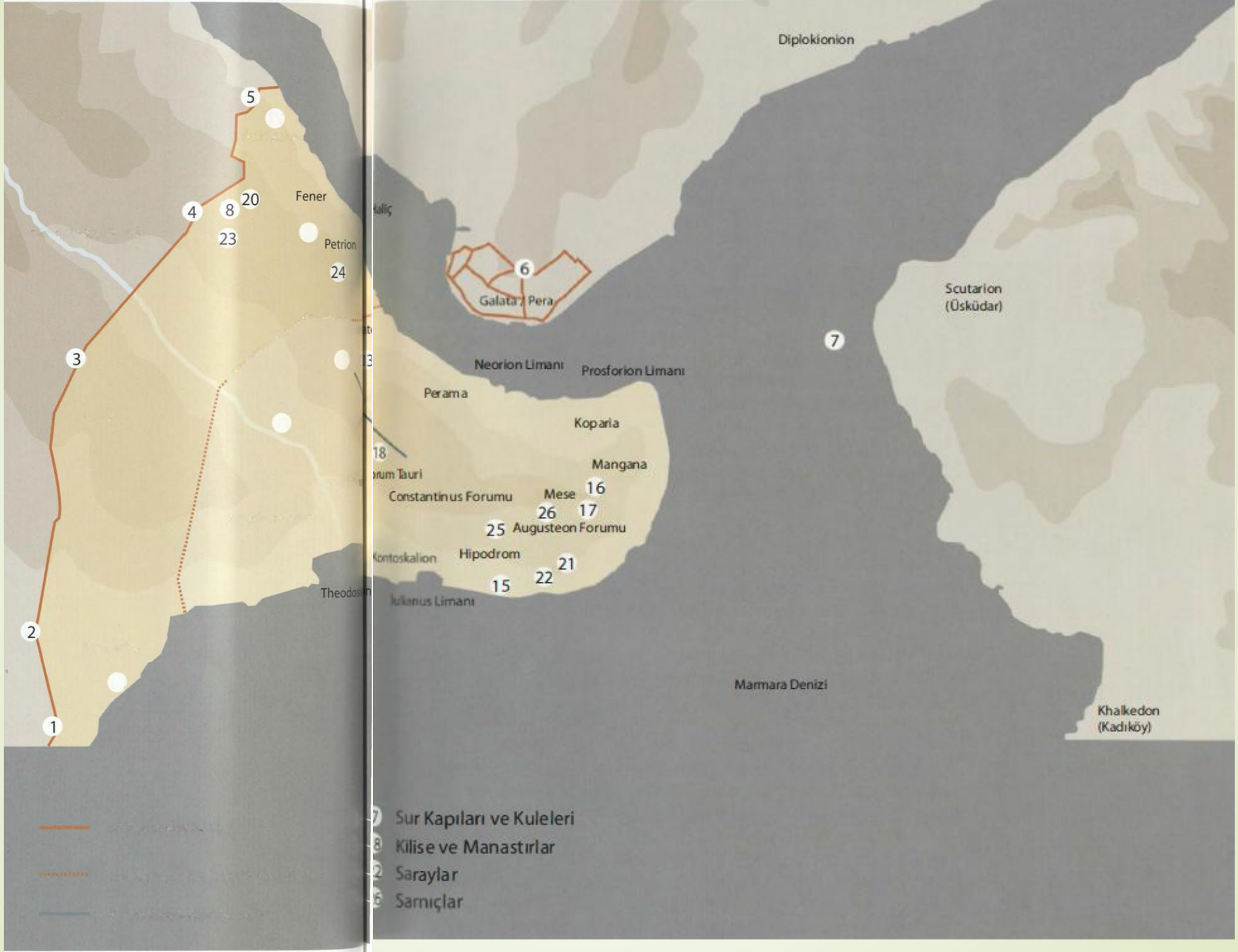


Week 4


Diplokionion
Fener
Petrion
Galata / Pera
Scutarion
(Üsküdar)
Neorion Limanı
Prosforion Limanı
Perama
Koparia
Mangana
Constantinus Forumu
Mese
Augusteon Forumu
Hipodrom
Kontoskalion
Theodosius Limanı
Marmara Denizi
Khalkedon
(Kadıköy)
7 Sur Kapıları ve Kuleleri
8 Kilise ve Manastırlar
2 Saraylar
6 Sarnıçlar

Architecture in Constantinople

(Roman and Byzantine period)

MILITARY ARCHITECTURE

-WALLS

Istanbul Walls:

- Land Walls
- Golden Horn Sea Walls
- Marmara Sea Walls

(It will not be reviewed here since it was discussed in sufficient detail in the last week.)

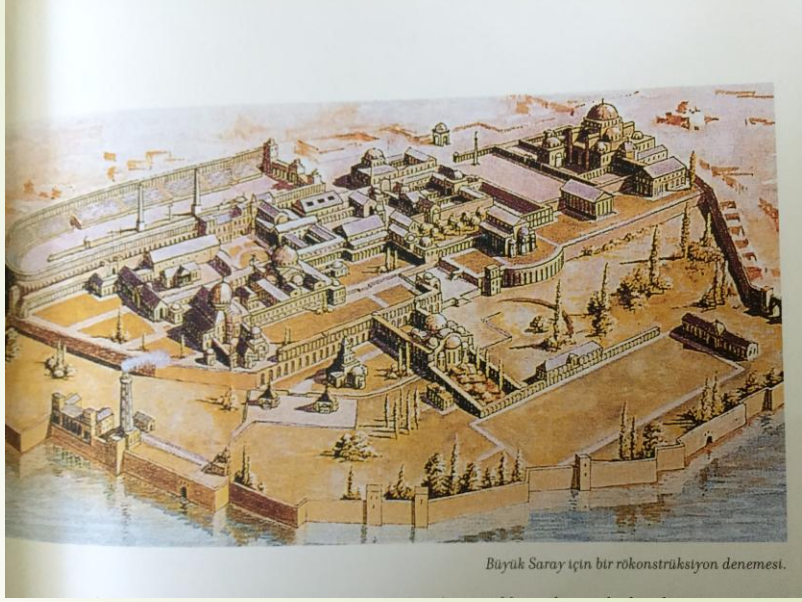


CIVIL ARCHITECTURE

- **Hippodrome** (It will not be reviewed here since it was discussed in sufficient detail in the previous weeks.)
- **Palaces**
- **Ports**
- **Water structures/Facilities**
  - aqueduct
  - cisterns
- **Square-obelisks**

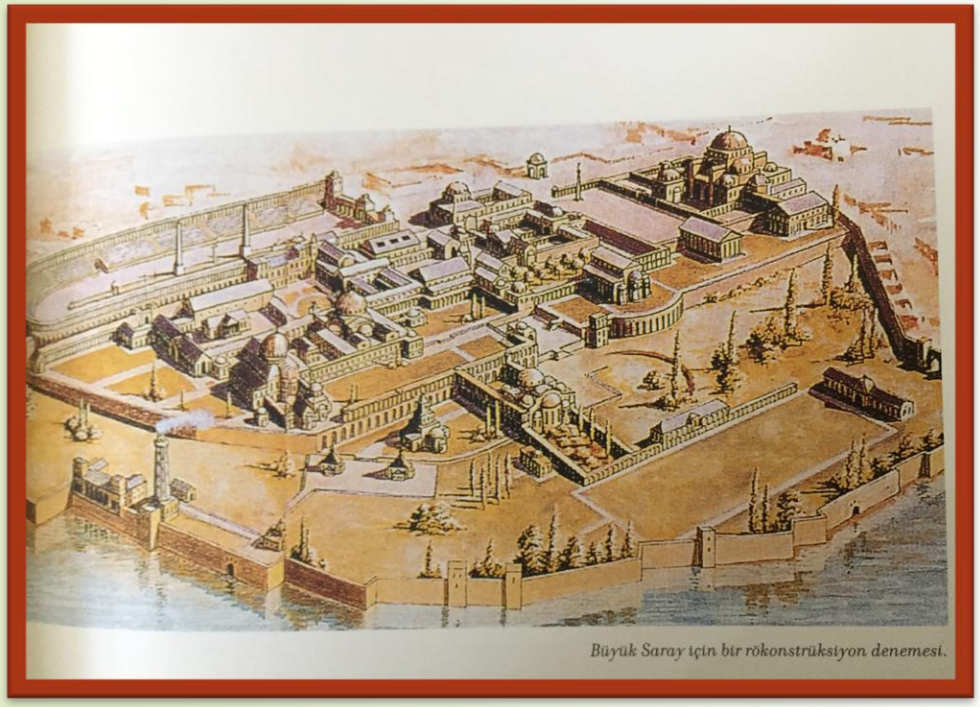
Palaces

- **The city, both inside and outside, was decorated with a set of imperial palace complexes**. In addition to the palaces belonging to the emperors, there were also **private palaces**. We have information about such structures from sources. Also, the architectural remains of some of them can still be seen.
- Some of **the summer palaces belonging to the Byzantine emperors** whose existence is known from sources include Magnaura and Secundiane in Hebdomon (Bakırköy); Rhegion (Küçükçekmece); Bryas (Küçükyalı); Damatris (Samandıra); Damalis (Üsküdar); Hieron (Fenerbahçe).
- **The oldest of the imperial palaces in the city is the structure known as the Great Palace.**
- The Great Palace remained in use between the **4th and 11th** centuries, and after this date, it was abandoned except for partial use except for special ceremonies because it required constant repairs.



A reconstruction attempt for the grand palace

Grand Palace (Magnum Palatium)



- The imperial palace, the Great Palace, covers an area of approximately 100,000 square meters from the Hagia Sophia - Hippodrome line to the sea.
- **A palace complex** surrounded by walls.
- **It contained throne and official reception halls, living areas, churches, baths, guard barracks, gardens and playgrounds.**
- **The first structures that formed the core of this palace were built by Constantine** on the eastern side of the Hippodrome, in the area where the Sultanahmet Mosque stands today.
- Later, some parts of the palace complex, which **expanded with various additions until the 11th century**, have survived to the present day. One of these is a mosaic-paved peristyle courtyard belonging to the later phases of the palace and unearthed during excavations between 1951-1954. These mosaics have been restored and are now open to visitors as the Mosaic Museum.

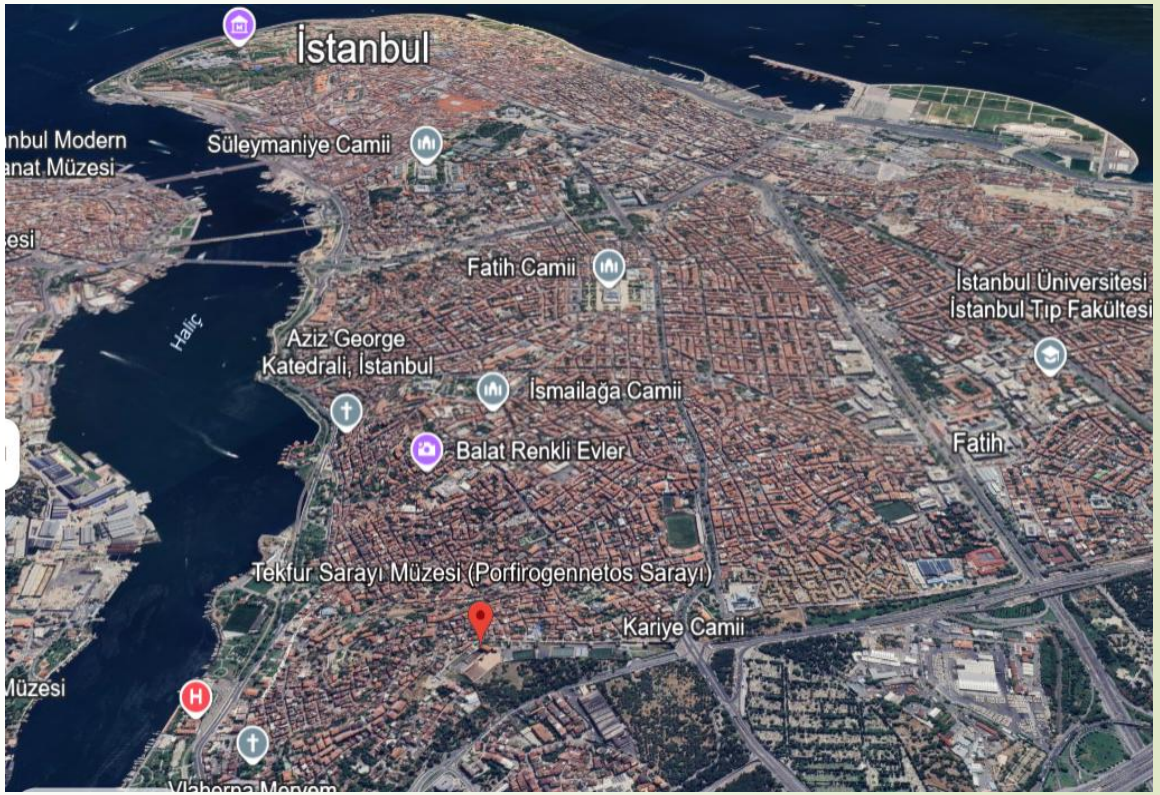




Mosaic samples from the grand palace

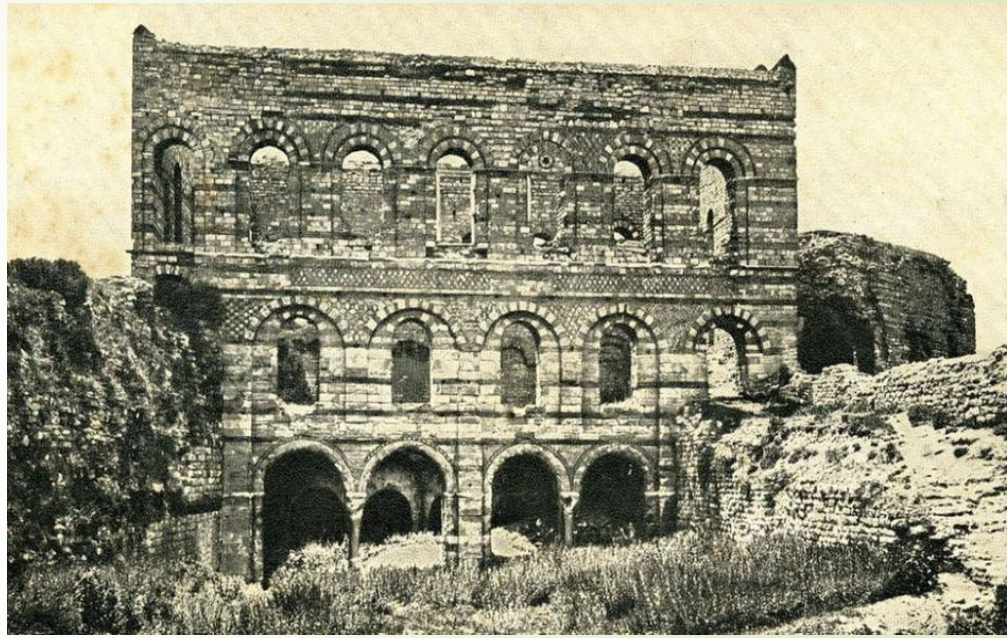
Palaces

- After the Great Palace was abandoned in the 11th century, **the Blachernae Palace**, located in today's Ayvansaray district, was used.
- The Palace of Blachernae was an imperial Roman residence in the suburb of Blachernae, located in the northwestern section of Constantinople (today located in the quarter of Ayvansaray in Fatih).
- **The only structure from the Blachernae Palace complex that remains standing today is the The Palace of Porphyrogenitus (Tekfur Palace)** rising at the southernmost end.



Palaces

- **The Palace of Porphyrogenitus (Tekfur Palace)** was among the palaces used by the emperors during the last glorious period of Byzantine palaces. It was in an ideal location for the protection of the emperors against a possible rebellion.
- During the Latin invasion of Istanbul that lasted 57 years, almost all of the Tekfur Palace and other imperial structures around it were burned and destroyed. Even the lead on the roof covering of the buildings was melted down and sold to Venetian merchants.
- After Fatih conquered Istanbul, it was repaired and used for various purposes. It fell into ruin again in the 17th century; some parts were used as an elephant stable and a zoo. Later, it was converted into a glass and tile workshop and became famous for its tiles. “Tekfur Palace tiles” were preferred in many architectural works.

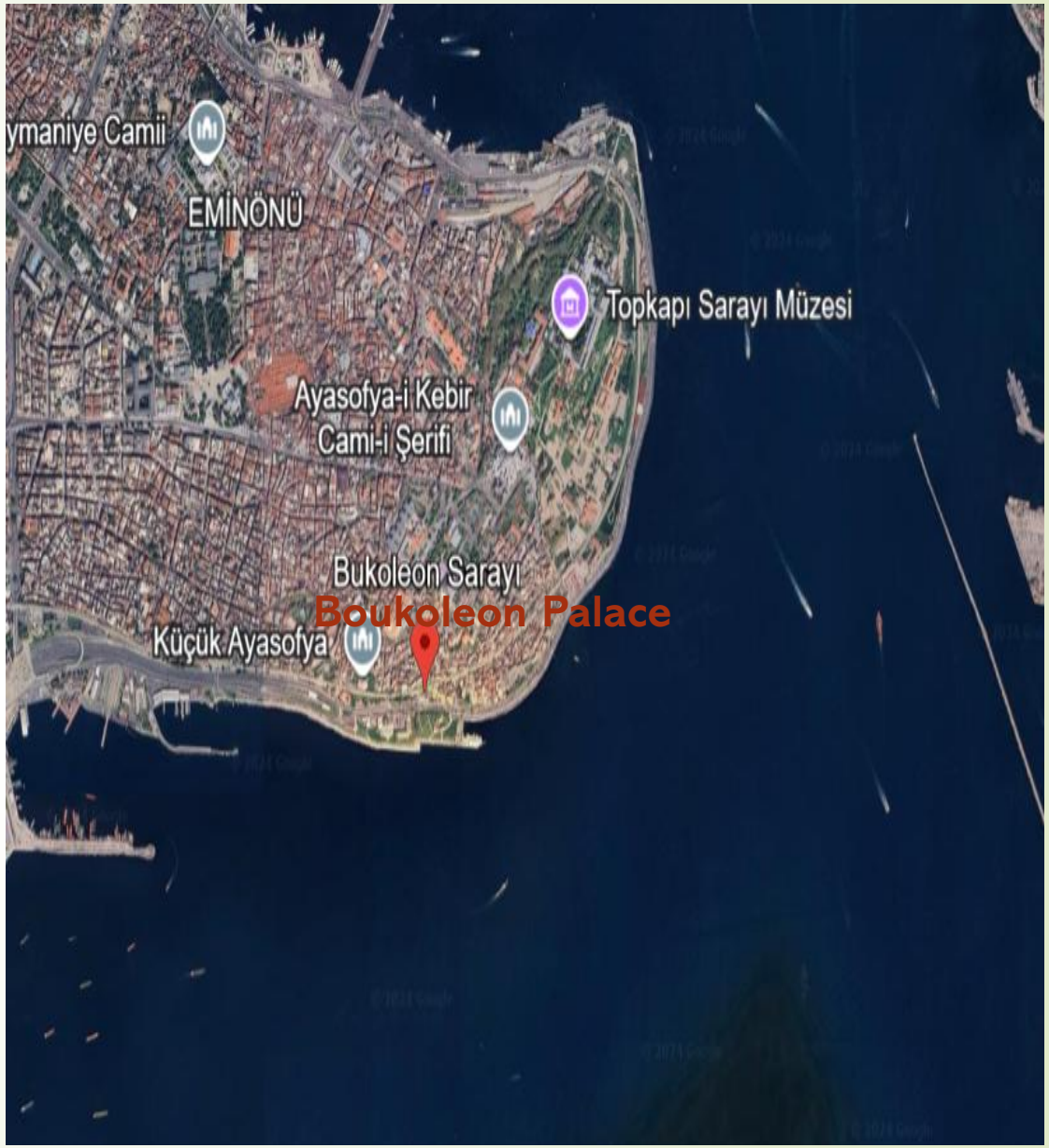


This building, which was also used as a glass and tile workshop during the Ottoman period, is today the **Tekfur Palace Museum**.

https://www.youtube.com/watch?v=eOmPZDCkgXw

Paleces

- The Blachernae Palace was destroyed during the Latin invasion in 1204; the new Emperor chose the better-preserved **Boukoleon Palace** as his residence.
- A part of the Great Palace, known as the Justinian House or Boukoleon Palace, still stands today near Ahırkapı.



Palaces

Bukeleon Palace

- It is assumed that the Boukoleon Palace and Port, integrated into the Sea Walls near Çatladı Kapı, took its name from the statues of a bull (bous) and a lion (leon) placed on two columns that once rose on the shore.
- It is thought that there was a palace structure belonging to Theodosius II (408-450) nearby, which was brought to its final form in the 10th century during the reign of Emperor Nikephoros II Phokas (963-969).
- Very few parts of the palace, most of which was destroyed during the construction of the railway in 1871, have survived to the present day.



https://www.youtube.com/watch?v=gAt4fg7NqrM

Palaces





Seaward Walls of Constantinople and remains of Boukoleon Palace (c.1850)



In the past, the coastline was far behind and the port and palace walls were very close.

Palaces

- Not far from the Grand Palace, between Ahırkapı and Sarayburnu, at the point where today is a military zone, was the **Mangana Palace.**
- The structure was created by demolishing the previously existing residences on the same site in the 11th century by imperial decree and **building a palace and monastery in their place**.
- After the conquest, the dilapidated superstructure was demolished during the construction of Topkapı Palace, while the substructure, which still exists today, was preserved.

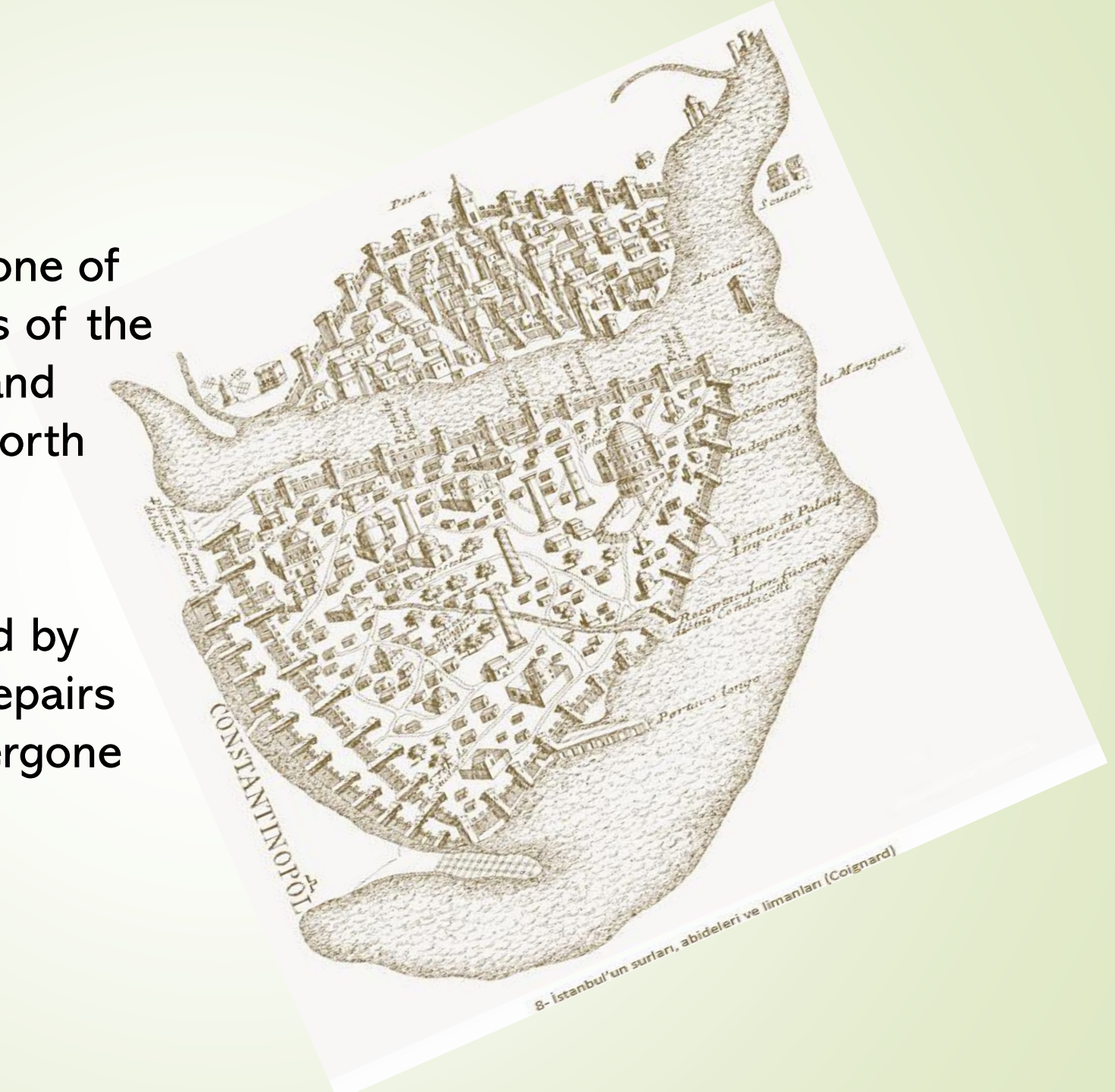
- https://www.thebyzantinelegacy.com/palace-mangana
- https://www.youtube.com/watch?v=QtKk15khcpY

Palaces

- In addition to the imperial palaces, there were also private palaces belonging to individuals in the city.
  - The most important of these are the Antiochus and Lausos palaces near the Binbirdirek Cistern in Sultanahmet, dating back to the 5th century, and the Princess Anicia Juliana Palace, located between the Istanbul Metropolitan Municipality building and the St. Polieuktos Church in the park area opposite it. The architectural remains of the Lausos Palace can still be seen today. A part of this palace is used today for theater and music activities. On the other hand, the architectural remains of the Anicia Juliana Palace, dating back to the 6th century, are not visible, while the floor mosaics from here are in the atrium of Hagia Irene.

Ports

- The ports, which constitute one of the most important elements of the city in terms of both **trade** and **military,** are located in the north and south of the historical peninsula.
- These ports have been called by different names due to the repairs and changes they have undergone over time.



8- İstanbul'un surları, abideleri ve limanları (Coignard)

Ports/Harbours

- Ports should not be evaluated as an architectural structure alone, but together with related structures. Accordingly, **the warehouses** built on the nearby shoreline and the breakwaters at the sea should also be considered as harbour installation. **Horreum**s, whose existence is known from sources and where substances such as grain and oil are stored, should not be far from the port site. However, no structure that has been identified as a horreum has been found in the city to date.
- Although there is no example that has survived to the present day in the form of a complete port, their locations are known.
- In particular, the recent excavations in Yenikapı have provided detailed information about the Port of Theodosios, one of the largest ports in the ancient world and the city.

Ports

Iulianos/Sofia/Kontoskalion/Kadirga Port

- **The earliest port in the southern part of the city** is the one named after Emperor Julianus, probably the expansion and arrangement of a natural bay where an earlier pier was located on the same site.
- After the great fire in 465, the interior of the port was deepened and expanded during the reign of Emperor Anastasius (491-518) and a breakwater was added.
- The port, which was destroyed after another fire in 561, was renovated this time by Emperor Justinus II (565-578) and from then on it was called the Port of Sofia, after the emperor's wife Sofia.
- The ports of Julianus and Sofia are the same. The Port of Julianus became the main port for wholesale trade in the 8th-9th centuries. It is thought that this port was later used as Kontoskalion and was probably one of the Byzantine naval bases.
- After the city fell into Byzantine hands again in 1261, the Port of Kontoskalion was rebuilt by Emperor Justinus VIII. It is known that it was turned into the main naval base of Byzantium by Michael Palaiologos (1259-1282). In 1462, renovation works were carried out under the name of Kadırga Harbor for the use of the Ottoman navy. **The harbor was filled by Grand Vizier Sokullu Mehmed** Pasha at the end of the 16th century because his palace in the same region was affected by the bad smells emitted by the harbor.

Ports

Theodosian Port

- A new port was established by **Emperor Theodosios** I at the end of the 4th century in a deeply indented bay at the point where the Likos (Bayrampaşa) Stream flows into the sea, and this port became known by the name of the emperor.
- Although archaeological excavations have been carried out in the port to date, its borders are not yet known. As understood from the ships found in the excavations, the use of the port continued until the 11th century.
- The port, which was abandoned after the 12th century, became a dumping ground for rubble.
- https://www.youtube.com/watch?v=FoTgTUOO7dw
- https://www.youtube.com/watch?v=eiqRfawIoWI (3.34 m.

Ports

Boukoleon Port

- It is known from sources that this port, whose existence is known but whose exact location cannot be determined, **functioned as the pier of the Grand Palace** rather than a commercial or military port directly.
- **The Boukoleon Port was directly connected to the Grand Palace in order to facilitate the emperor's travels by sea**, especially within the city.
- **The small port, which was only allowed to be used by imperial ships**, was destroyed by the destruction that began with the construction of the railway in 1871, the fire on June 3, 1912, and the coastal road that was started to be built in 1957.

Ports

Port of Neorion

- This facility, which is located near today's New Mosque, is Neorion, known with its small square of the same name, from the Golden Horn ports known to have existed since the city's foundation.
- It is thought that the port is near Eminönü New Mosque.

Port of Prosoforion

- It is thought that the Prosoforion Port was located in today's Sirkeci.
- It is known that the port was used for a long time, as Venetian merchants were given permission to trade grain using this port in the 15th century.

• Water Facilities: ---Aqueducts

- **The first aqueducts in the city were built in the 2nd century during the reign of Emperor Hadrian (117-138).** There is no aqueduct that can be dated to the period of Constantine I or his successors.
- On the other hand, **the aqueduct, whose remains can still be seen today and which is named after Emperor Valens (364-378) who had it built, was completed in 373.**
- This aqueduct, which was partially destroyed by the Avar attacks in 626, continued to be used, albeit on a small scale, and underwent extensive repairs in 767. The latest repair records regarding aqueducts and their facilities date back to the first half of the 11th century.
- Although a 10th century source mentions the existence of an aqueduct in today's Cerrahpaşa district, which was known as Xerolophos (Dry Hill) during the Byzantine period because it was one of the regions with the least water in the city, it is not possible to find any trace of it.
- It is possible to come across traces of Byzantine architecture, albeit small, in the aqueducts of Büyük Bent, Kovuk, Uzun, Mazul, Kara, Viran, Veysi Baba and Kahveci, which were part of the systems called Kırkçeşme and Halkalı waters during the Ottoman period. Based on these and the aqueducts in Thrace that have survived in much longer parts, it is possible to say that **the city's water needs were mainly provided from the Thrace region and the west of the city during the Byzantine period.**

Valens/ Bozdogan Aqueduct

- Built by Emperor Valens before the Roman Empire was divided into East and West, the aqueduct was used to solve the city's water problem during the Byzantine and Ottoman periods after Rome.
- A significant portion of Istanbul's water needs were met by this aqueduct for approximately 1500 years.
- It has an average height of 28 meters and a length of 970 meters.
- The aqueduct was built so that water brought from water basins outside the city could cross the valley between the third and fourth hills of Istanbul.
- Pipes were laid in the open water channels on the aqueduct in later years.
- The structure, which is **the oldest aqueduct in Istanbul**, is today known as the Bozdoğan Aqueduct.



https://www.youtube.com/watch?v=7pGlRuoyQpM

Water Facilities
---Cisterns

- Initially very large, the cisterns were later transformed into much smaller ones, and sometimes by giving new functions to structures that originally had different functions.
- As of today, there are more than 100 cisterns within the city walls of Istanbul.
- They can be divided into two groups: **Open Air Cisterns** and **Covered Cisterns**.



Open Air Cisterns

- Open-air cisterns, similar to those found in the Roman Empire, attract attention with their giant size.
- One of these cisterns is the Fildamı Cistern, which is located outside the city walls in Veliefendi/Bakırköy and has dimensions of 126x86x9 m.
- There are three of them within the city walls. Each of these cisterns is located on a hill and is built with alternating stone and brick. There are traces, albeit few, of hydraulic plaster being used to prevent the leakage of the huge amounts of water they hold.



Fildamı Cistern

Open Air Cisterns

- These cisterns, which lost their original function over time, were used as vegetable gardens during the Ottoman period and were called **Çukurbostan** among the public because they were located at a lower level than the surrounding buildings and streets.
- They were named after the neighborhoods they were located in: Karagümrük Çukurbostanı, Çarşamba Çukurbostanı, Altı mermer Çukurbostanı. Although it is difficult to determine exactly when these cisterns were out of use, it is possible to say that they probably acquired a new function in the late Byzantine period and were converted into agricultural land. The absence of traces of materials and repairs from the years after they were built supports this claim.

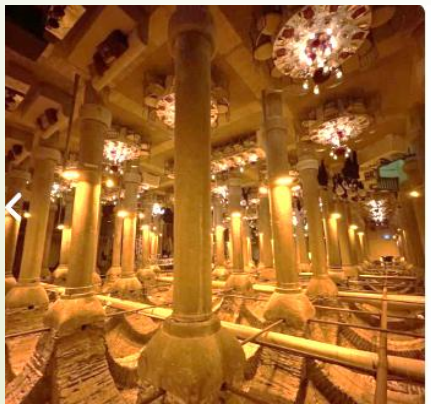
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Covered Cisterns

- **The largest covered cisterns are especially concentrated around the current Sultanahmet district**, which forms the city center.
- They were built to provide support when the open-air cisterns built about a century ago were inadequate.
- The hydraulic plaster and the rounding of the walls at the skirting level, which can be observed in very small sections in open-air cisterns, can be observed in almost all of the covered cisterns.
- The floors of the covered cisterns were mainly covered with bricks.
- It is thought that **the Binbirdirek Cistern** (63x56x9 m in size), **one of the largest covered cisterns in this region that provided water to the Great Palace and the surrounding structures, was built by Philoxenos around 525.**

Binbirdirek Cistern/ Cistern of Philoxenos

- The cistern has a plan shape of 64 X 56 meters and has an area of more than 3000 square meters.
- It had 224 marble columns in 16 rows. However, 212 of them have survived to the present day.
- In Istanbul, we only come across carriers consisting of two bodies superimposed on each other in this structure. A circular marble bracelet protruding outward was placed between the two superimposed columns and lead sheets were placed at the joints. Thus, the depth of the cistern was increased and its capacity to store forty thousand cubic meters of water was reached.
- In the Ottoman period, since there was no water in it anymore, silk and yarn spinners started working in the cistern. It is said that the name Binbirdirek was given at that time because of the seemingly countless columns in the darkness.
- During the reign of Murat IV, the Tayyarzade and later the Fazlı Paşa mansions were built on it..






- After the Binbirdirek Cistern, the construction of the structure known today as **the Basilica Cistern** (135x68x8 m) was completed.
- According to the statement of the 6th century historian Prokopios, the construction of this structure became a necessity after a dry summers. It was partially demolished for the construction of the open courtyard of the basilica located in the same place, and for this reason, **the structure known as Basilika Kisterna (Basilica Cistern) was completed in the early 530s by the order of Emperor Justinian I.**
- In order to save time and money, **column bodies, bases and capitals brought from different structures were used in the building**.
- 336 columns were used in the building measuring 140x70x11 m.

Basilica Cistern/ Tr. Yerebatan Sarnıcı

- **The Basilica Cistern, which is the largest covered cistern in the city with a water storage capacity of 80,000 tons** and draws attention with having **more reused load-bearing elements** than other covered cisterns, is a giant rectangular structure covering an area of approximately 10,000 m2.
- According to written sources, the Basilica Cistern provided water to the city from the Hadrianus Water Supply Line, one of the historical waterways. It distributed water to the Great Palace where the emperors resided and the surrounding structures for hundreds of years, also
- It is also known that the historical cistern, which was used for a while for the needs of Topkapı Palace after the conquest of Istanbul in 1453, was used by the public as housing gradually began to develop in the region.



Basilica Cistern

- **The most important symbol of the Basilica is Medusa head**. It is a relief blocks. It was used as a column base is upside down on the west side of the structure, while the one on the east is horizontal. Being as one of the most special examples of Roman sculpture, Medusa heads have also been the subject of many legends:
- According to one legend, Medusa is one of the three Gorgons, the female monsters of the underworld in Greek mythology. Of these three sisters, the snake-headed Medusa has the power to turn those who look at her into stone.
- According to another rumor, Medusa is a girl who boasts about her black eyes, long hair and beautiful body. Medusa loves Perseus, the son of Zeus. Meanwhile, Athena loves Perseus and is jealous of Medusa. For this reason, Athena turns Medusa's hair into snakes. From now on, everyone Medusa looks at will turn into stone. Later, Perseus cuts off Medusa's head and defeats many of his enemies by taking advantage of this power.



Spaces Transformed into Cisterns

- The most important of the places that were originally used for another purpose but were later converted into cisterns is the one in the Hippodrome.
- **The southern part of the Hippodrome**, which was rarely used for its original purpose from the 11th century onwards, was converted into a cistern. It is not known exactly when this conversion took place.
- Another example is the cistern of **the Mirelaion (Bodrum Mosque) Monastery.** This place, thought to date back to the 5th century, was converted into a cistern during the construction of the monastery in the 10th century.

https://www.youtube.com/watch?v=M-Jt2Rd1wF8



Bodrum Camii

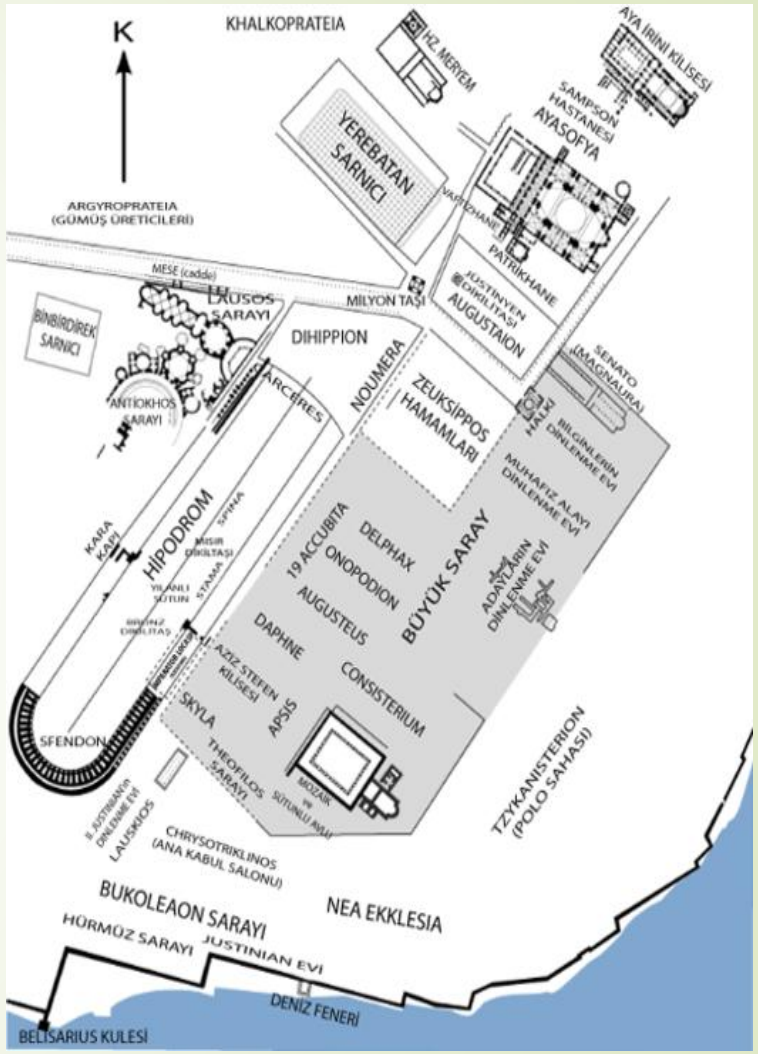
The structure is located at south of Ordu Street in Lâleli. A monastery was founded by Romanos I Lekapenos in the 10th century. There is a water cistern and a covered bazaar under the structure. It was built as part of the Myrelaion monastery next to the Myrelaion church. After the conquest of Istanbul, it was converted into a mosque by Grand Vizier Mesih Pasha during the reign of Bayezid II.

Squares-Obelisks

- The city has a different planning than its predecessor, Rome. There is no question of reconstruction activities carried out in Rome, especially after the great fires. For this reason, the city took shape with additions made according to need.
- While the existence of some of the squares in the city is known from sources, some should be evaluated as obelisks and the small squares that should have been around them.

Squares-Obelisks

- The most important square in the city, **Augustaion**, is today's Hagia Sophia Square, based on the title Augusta, the Latin equivalent of empress, for Helena, mother of Constantine I (324-337).
- Although its borders are not known exactly, to the west there must be the Mese road, to the east there must be the garden of the Four Seasons Hotel, where the senate building is located, and to the south there must be the Zeuksippos Baths (today's fountain park).
- **The use of this square continued in the following centuries, and Emperor Justinian I** (527-565) **had an equestrian statue of himself placed on top of a column named after him**.
- It is learned from the writings of Paspates, a local of the city, that this statue was still visible at the end of the 19th century.





Egyptian obelisk found in the Hippodrome

- In the middle of the hippodrome, there was a section called the spina, which divided the race track into two. The race cars would complete the race by going around the spina a few times.
- Today, **the location of the spina is marked by three monuments on it:** The first of these is **the Egyptian obelisk.**
- The other monument is **the serpent column** on the south side of the obelisk.
- The third monument of the spina that has survived to the present day is the **Walled obelisk** on the sphendone side, built in the tenth century.



the serpent column and The Egyptian obelisk



The Egyptian obelisk and the Walled obelisk

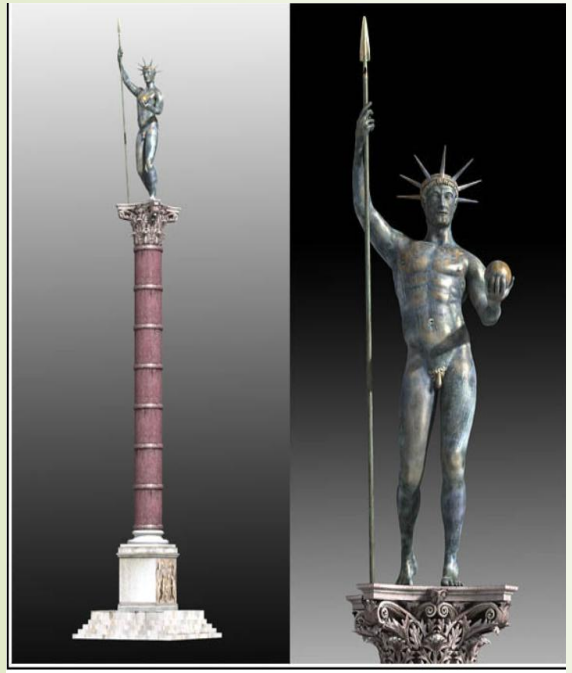
3 monuments of the Spina in the Hippodrome



Hipodrom ve Büyük Saray'ın duvarları (Ön tarafta)

The Mese Street opened to an oval-shaped forum named after Constantine in today's Çemberlitaş, with an obelisk in the middle, and then continued westward.

- On the monument-column made of prophyry granite (Çemberlitaş) in the middle of the forum, there was a statue of Constantine resembling the sun god Apollo.
- It is known that the streets and squares of the city of Constantinople were decorated with many statues.



This reconstruction shows the Column of Constantine before 1106.
This reconstruction is created for "Jonathan Bardill, Constantine: Divine Emperor of the Christian Golden Age (Cambridge University Press, forthcoming)".

Constantine Forum
and
Constantine Column/Çemberlitaş Column



The Column of the Goths (Gotlar Sütunu)

- It is located at one of the points closest to Gülhane Park and Topkapı Palace.
- It is called by this name because of the Emperor Claudius Gothicus mentioned in the inscription on it.
- It is known that the Goths attacked the city between 258-269 and were finally defeated by Emperor Claudius II. It is thought that the column was erected in memory of this event.



Forum Tauri- Forum of Theodosius

- The exact construction date of Forum Tauri-Forum Theodosios, parts of which are located opposite and next to the Istanbul University Faculty of Letters building between Beyazıt and Laleli, is unknown, but it is known that it was built during the reign of Constantine I or later Valens, and reorganized during the reign of Theodosios I and opened in 393.
- Although the reason why the square was given the name Tauri is not known for sure, it has been claimed that it was based on the name of a mayor or because of a bull statue (tauros = bull) found here.
- Nothing remains today from the column in the middle of the square with an equestrian statue of Emperor Theodosios on it.
- On the other hand, parts of a three-arched arch at the entrance to the forum can be seen both on the roadside and in the lower sections of Beyazıt Bath, where they were used as reused building material.

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Theodosius Forumunun,

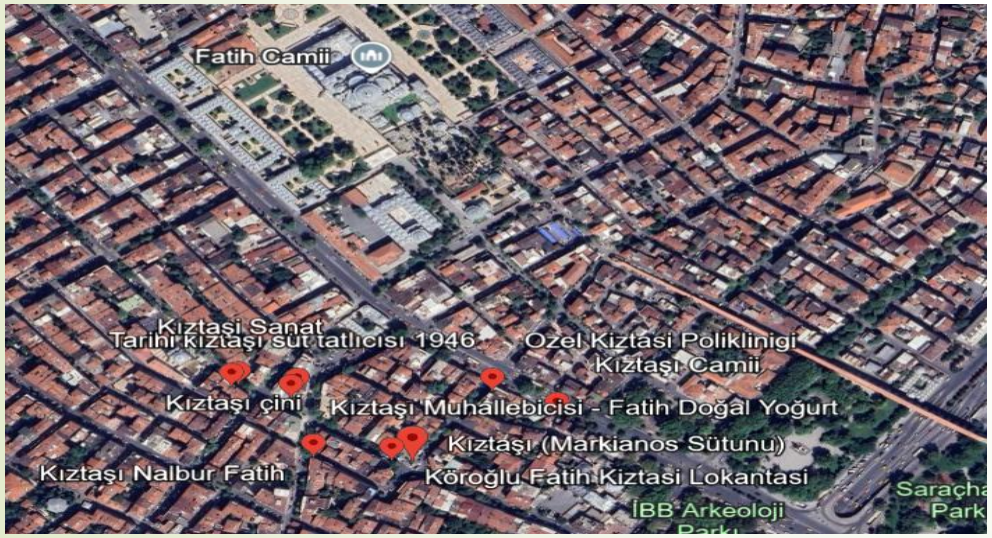
Ruins of the forum

Arcadius Forum

- Only the base of the column with the same name remains from the Arkadios Forum in the Cerrahpaşa district.
- It is known that the square and the column with the statue of Emperor Arkadios in the middle were completed by his son, Emperor Theodosios II. The date of completion of the square is thought to be 435.
- The statue on top of the column, which was damaged by earthquakes in the 6th century, fell in 740 and was not replaced.

Maiden's Stone/Markianos Column (Kıztaşı)

- Another column, surrounded by a small square as it is today, was erected around 450 by Governor Tatiatus in the name of Emperor Marcianus, and was called the Maiden's Stone during the Ottoman period because of the depictions of Nike bringing victory on it.





https://www.youtube.com/watch?v=uX4UJv-eIjQ